



UNIVERSITAS
GADJAH MADA

Penerapan Single Layer Perceptron dalam Spreadsheet dan Kode Python [REV]

Tugas 1 Pembelajaran Mesin Mendalam

Oleh: DHIMAS PUTRA SULISTIO (24/537952/PA/22811)

Google Sheet: Accuracy of Training and Validation

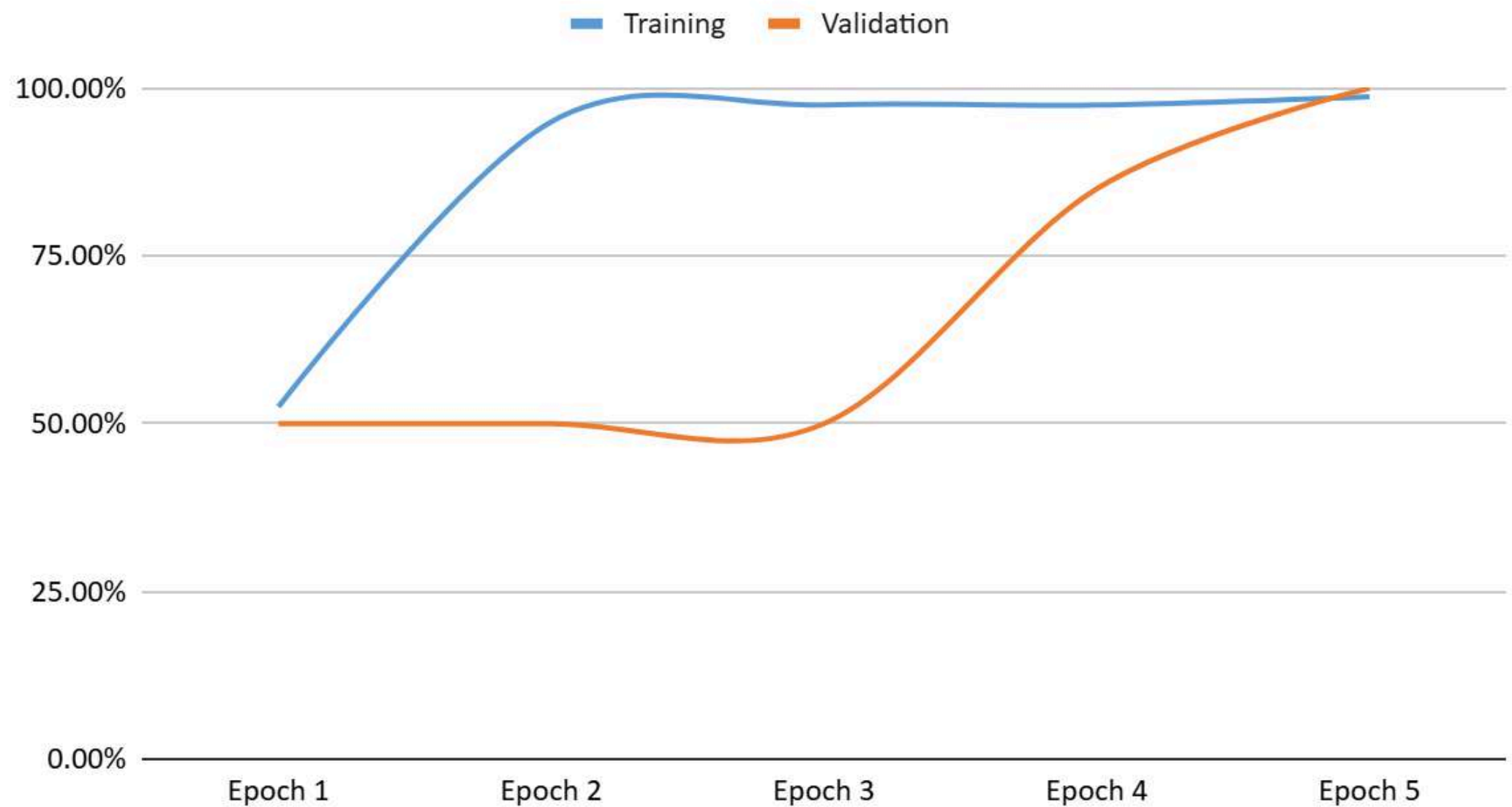
Data Training

AKURASI		
Epoch	Jumlah Sesuai Target	Persentase
Epoch 1	42	52.50%
Epoch 2	76	95.00%
Epoch 3	78	97.50%
Epoch 4	78	97.50%
Epoch 5	79	98.75%

Data Validation

AKURASI		
Epoch	Jumlah Sesuai Target	Persentase
Epoch 1	10	50.00%
Epoch 2	10	50.00%
Epoch 3	10	50.00%
Epoch 4	17	85.00%
Epoch 5	20	100.00%

Akurasi Training VS Validation



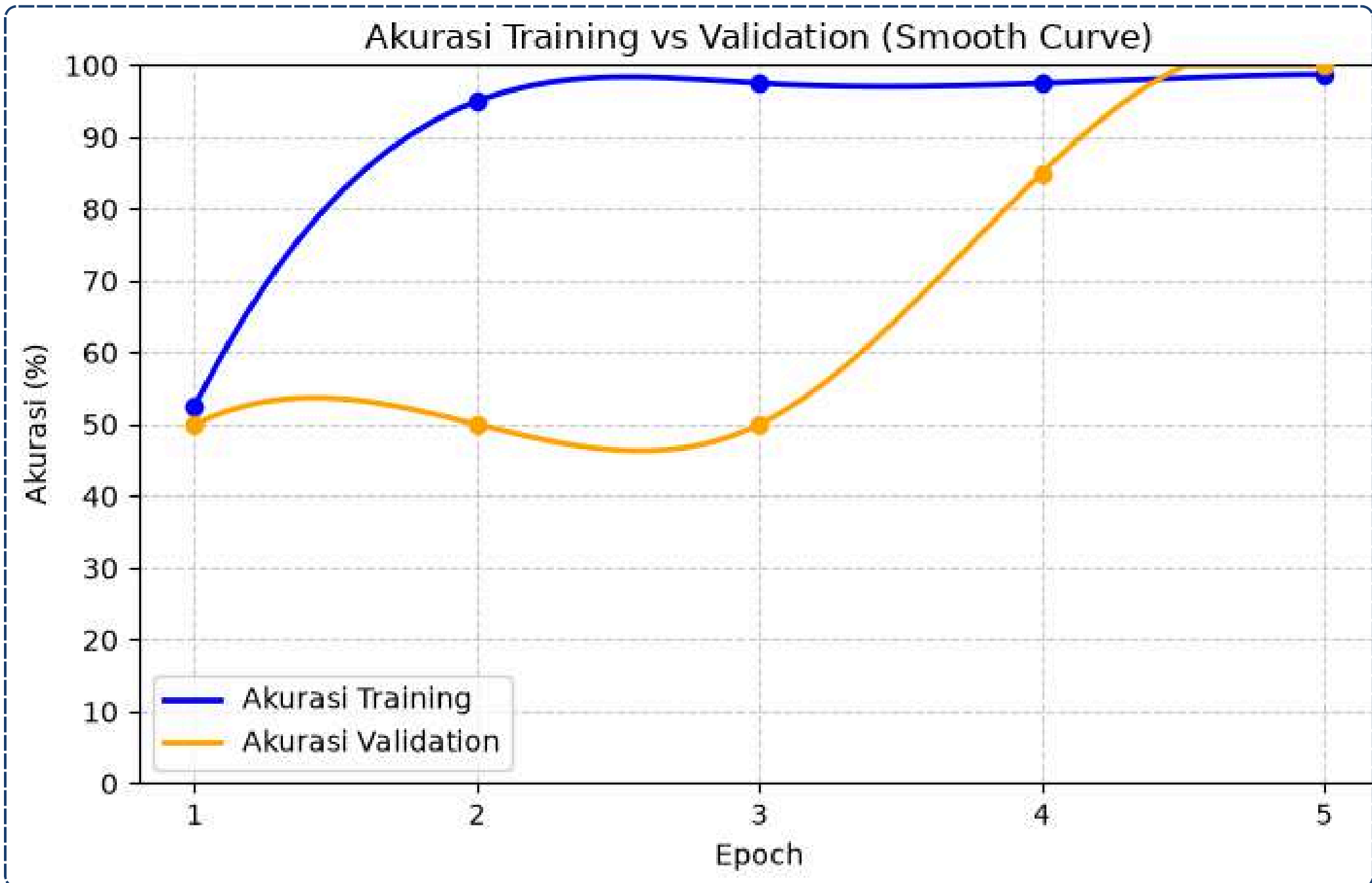
Python Code: Accuracy of Training and Validation

Data Training

```
Epoch 1 : 52.5 %  
Epoch 2 : 95.0 %  
Epoch 3 : 97.5 %  
Epoch 4 : 97.5 %  
Epoch 5 : 98.75 %
```

Data Validation

```
Epoch 1 : 50.0 %  
Epoch 2 : 50.0 %  
Epoch 3 : 50.0 %  
Epoch 4 : 85.0 %  
Epoch 5 : 100.0 %
```



Google Sheet: MSE of Training and Validation

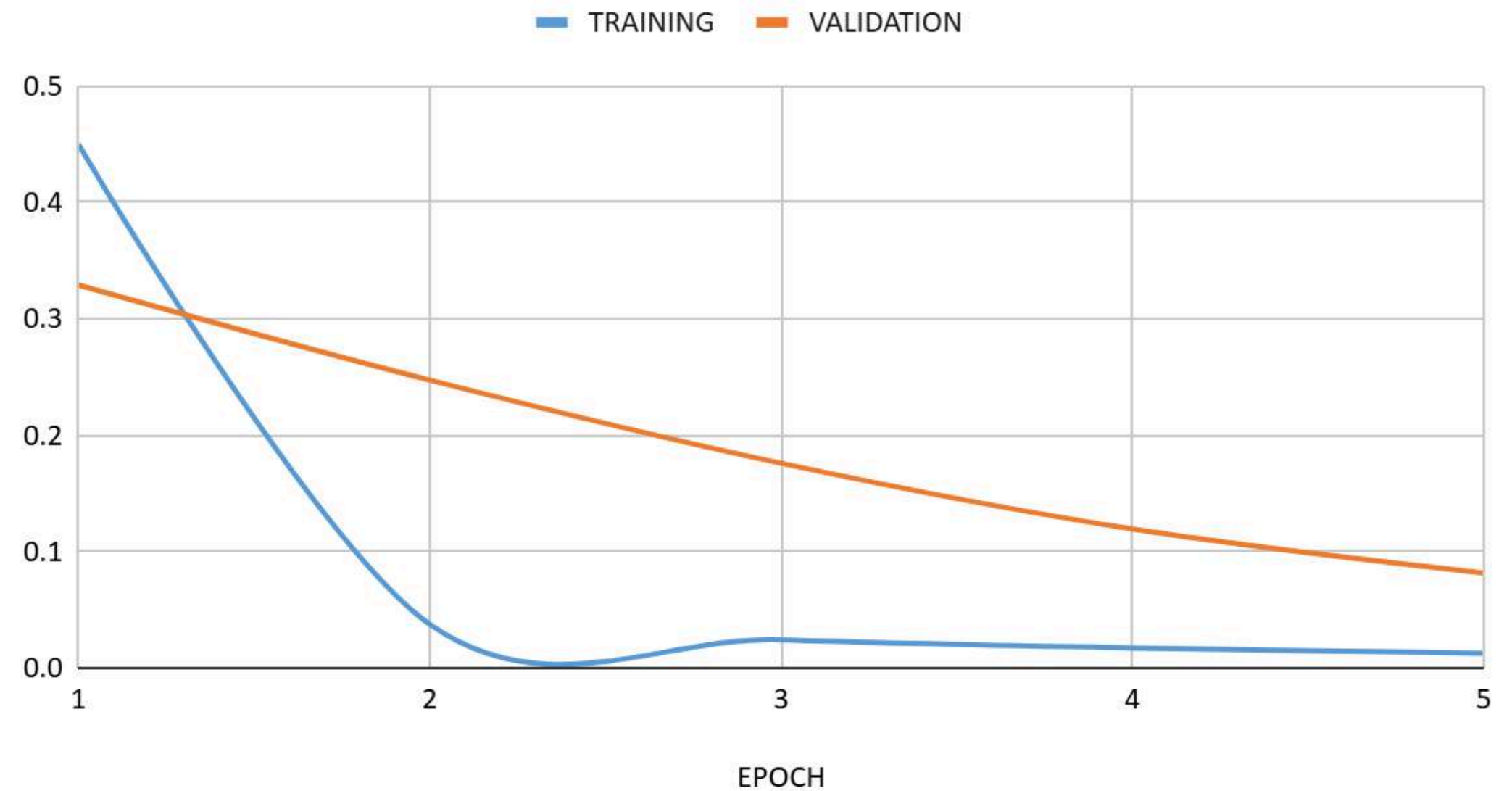
Data Training

Epoch	MSE
1	0.4498886551
2	0.03745191423
3	0.024372195
4	0.01735720168
5	0.01274015821

Data Validation

Epoch	MSE
1	0.3289511566
2	0.2472893233
3	0.1758916407
4	0.1193813733
5	0.08158136941

Lost Function TRAINING VS VALIDATION



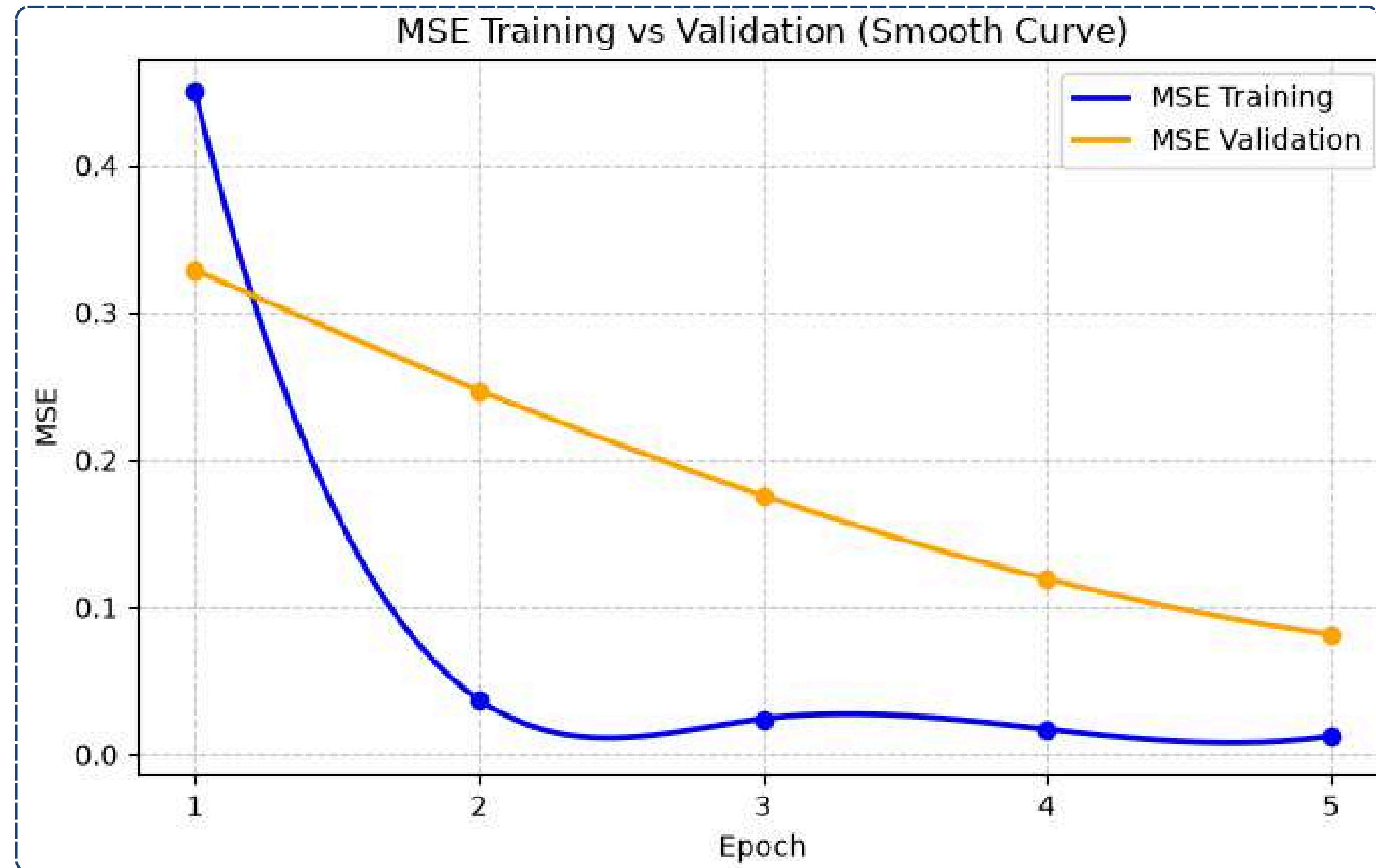
Python Code: Accuracy of Training and Validation

Data Training

```
=====MSE TRAINING=====
Epoch 1 : 0.4498886551395292
Epoch 2 : 0.0374519142280995
Epoch 3 : 0.02437219499639377
Epoch 4 : 0.0173572016784266
Epoch 5 : 0.012740158208926562
```

Data Validation

```
=====MSE VALIDATION=====
Epoch 1 : 0.3289511565546991
Epoch 2 : 0.24728932330356232
Epoch 3 : 0.17589164066713475
Epoch 4 : 0.11938137329297324
Epoch 5 : 0.08158136941106596
```



Source URL

Google Sheets URL: <https://docs.google.com/spreadsheets/d/1PYp3IYPpWIdSuQa2Yd0xuSloeYpO-JpK/edit?usp=sharing&ouid=107964778106247645635&rtpof=true&sd=true>

Github Repo URL: <https://github.com/muddglobb/DEEP-LEARNING>

References

<https://matplotlib.org/stable/tutorials/pyplot.html>



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Terima Kasih.

*Merakyat, Mandiri, Berkelanjutan
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